



SIMATS
ENGINEERING



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Compiler Construction Tools and LEX

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Introduction

- A compiler translates high-level language into machine code.
- It checks program errors before execution.
- It improves execution speed.
- It converts source code into executable code.

Phases of Compiler

- Lexical Analysis – Converts characters into tokens.
- Syntax Analysis – Checks grammar.
- Semantic Analysis – Checks meaning.
- Intermediate Code Generation.
- Code Optimization.
- Code Generation.

Compiler Construction Tools

- LEX/Flex – Generates lexical analyzers.
- YACC(Yet Another Compiler Compiler)/Bison – Generates parsers.
- ANTLR(Another Tool for Language Recognition) – Parser generator.
- LLVM (Low Level Virtual Machine)– Compiler infrastructure.
- GCC(GNU Compiler Collection) – Compiles programs into machine code.

LEX – Introduction

- LEX is a lexical analyzer generator.
- Developed for Unix systems.
- Uses regular expressions.
- Produces C source code for scanners.
- Works with YACC.

Structure of LEX Program

- Definition Section – Header files and declarations.
- Rules Section – Regular expressions and actions.
- User Code Section – Additional C functions.
- Sections are separated by %% symbols.

Working of LEX

- Source Program
- ↓
- LEX Scanner
- ↓
- Pattern Matching
- ↓
- Token Generation
- ↓
- Parser (YACC)
- ↓
- Syntax Tree

Regular Expressions

- Identifier: `[a-zA-Z_][a-zA-Z0-9_]*`
- Number: `[0-9]+`
- Keyword: `if|else|while`
- Operator: `+ - * /`
- White spaces are ignored.

Advantages & Applications

- **Advantages:**

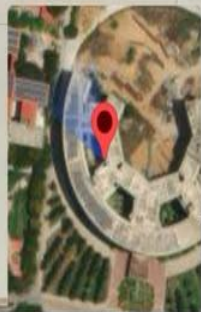
- • Fast token generation
- • Easy maintenance
- • Reduces manual coding

- **Applications:**

- • Compilers
- • Interpreters
- • IDEs
- • Syntax highlighting

Conclusion

- LEX simplifies lexical analysis.
- It improves compiler efficiency.
- Widely used with parser generators.



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